Classifiers and Confidence Estimation for Oil Spill Detection in ENVISAT ASAR Images

Camilla Brekke Anne H. S. Solberg

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Abstract

This paper proposes an improved classification framework for automatic oil spill detection in ENVISAT ASAR images. The approach focuses on the classification stage of an oil spill detection system and compares the performance of regularized statistical classifiers and support vector machines (SVMs). To address the tradeoff between true detections and false alarms, an automatic confidence estimation scheme is introduced. Experimental results demonstrate that regularized statistical classifiers outperform SVMs for this problem and that confidence estimation provides valuable operational support.

1 Introduction

Oil spills appear as dark areas in synthetic aperture radar (SAR) images because oil dampens short gravity waves on the sea surface. A major challenge in oil spill detection is distinguishing true oil slicks from naturally occurring low-backscatter ocean phenomena known as look-alikes.

The general oil spill detection framework consists of dark spot detection, feature extraction, and classification. While segmentation and feature extraction have been studied extensively in earlier work, this paper focuses on improving the classification step.

Oil spill detection is a rare-event problem where look-alikes occur far more frequently than true oil spills. Therefore, minimizing false alarms while maintaining high detection probability is critical for operational use.

2 SAR Images and Dataset

The system is trained and evaluated using 103 ENVISAT ASAR Wide Swath Mode images acquired over the Baltic Sea and North Sea between 2003 and 2005. The dataset is divided into training, validation, and test sets.

Aircraft verification data are available for the test set, providing reliable ground truth. Oil slick candidates in the training and validation sets were manually labeled with different confidence levels, which are later used in training the confidence estimation module.

3 Classification Methodology

For each detected dark spot, a feature vector is computed. The features describe slick shape, contrast, texture, and surroundings. Shape features include slick area, width, complexity, moment-based descriptors, and angular measures. Contrast and texture features include local contrast, border gradient, smoothness contrast, power-to-mean ratio, and variance.

A new binary feature indicating low-wind areas is introduced to improve discrimination between oil slicks and look-alikes caused by calm sea conditions.

The classification problem is addressed using both statistical classifiers and support vector machines.

4 Statistical Classifier

The statistical classifier is based on Bayes' theorem, combining prior probabilities with class-conditional feature densities. Oil spills and look-alikes are modeled using multivariate Gaussian distributions.

Because feature behavior varies with wind conditions, the dataset is divided into subclasses based on wind level and shape descriptors. This subdivision produces more homogeneous feature distributions within each subclass.

To handle severe class imbalance and limited training data, covariance matrix regularization is applied. The regularized covariance matrix is formed as a weighted combination of the full covariance matrix and its diagonal version. The regularization parameter is selected using cross-validation.

Loss functions are introduced to penalize misclassification of oil spills more heavily than misclassification of look-alikes.

5 Support Vector Machines

Support vector machines are evaluated as an alternative classifier. A C-support vector classifier with a radial basis function kernel is used. Due to class imbalance, different penalty parameters are assigned to oil spills and look-alikes.

Grid search and cross-validation are used to select optimal kernel and penalty parameters. Feature scaling is applied prior to training.

Despite these adaptations, SVM performance is found to be inferior to that of the regularized statistical classifier for this application.

6 Automatic Confidence Estimation

To support operational decision-making, an automatic confidence estimator is introduced. Each slick classified as an oil spill is assigned one of four confidence levels: High, Medium, Low, or Very Low.

Confidence estimation is based on a rule-based system that combines selected features such as local contrast, homogeneity of surroundings, distance to ships or oil rigs, slick area, and shape descriptors. Feature thresholds are estimated semi-automatically from training data and refined through validation.

The confidence levels provide an intuitive measure of reliability and help operators prioritize alarms.

7 Methodology

The complete two-step methodology can be summarized as follows:

1. Detect dark spots in ENVISAT ASAR images.
2. Extract geometric, radiometric, texture, and contextual features.
3. Divide samples into subclasses based on wind level and shape descriptors.

4. Train a regularized statistical classifier using Bayesian decision theory.
5. Classify dark spots as oil spills or look-alikes.
6. Assign confidence levels to detected oil spills using a rule-based confidence estimator.

This two-step approach mimics expert human reasoning while remaining computationally efficient.

8 Performance Comparison

Performance evaluation compares the regularized statistical classifier, a nonregularized statistical classifier, and SVMs. The regularized statistical classifier significantly reduces false alarms while improving oil spill detection rates.

Compared to earlier classifiers, the regularized approach reduces false positives by several hundred detections while losing fewer true oil spills. SVMs produce higher false alarm rates and lower detection accuracy.

The introduction of confidence levels further improves usability by allowing operators to trade off detection sensitivity against false alarm rate.

9 Discussion and Conclusion

This paper demonstrates that regularization of covariance matrices substantially improves statistical classification for oil spill detection in SAR imagery. Compared to SVMs, the regularized statistical classifier offers superior robustness for imbalanced datasets.

The automatic confidence estimation provides valuable operational support by ranking detected slicks according to reliability. The proposed framework reduces manual workload while maintaining high detection accuracy.

Future work includes incorporating additional environmental information such as wind and biological phenomena to further improve classification reliability.